TC and SC Practice Questions

Find the Time Complexity (TC) and Space Complexity (SC) for each question.

Question 1

```
int sum = 0;
for(int i = 0; i < n; i++) {
sum += arr[i];
}
```

Question 2

```
for(int i = 0; i < n; i++) {
System.out.println(i);
}
for(int j = 0; j < n; j++) {
System.out.println(j);
}
```

Question 3

```
for(int i = 1; i < n; i *= 2) {
count++;
}
```

Question 4

```
for(int i = n; i > 0; i /= 2) {
System.out.println(i);
}
```

Question 5

```
int[][] matrix = new int[n][n];
```

Question 6

```
void fun(int n) {
if(n == 0) return;
fun(n - 1);
}
```

Question 7

```
for(int i = 0; i < n; i++) {
for(int j = 0; j < n; j++) {
System.out.println(i + j);
}
}
```

Question 8

```
for(int i = 0; i < n; i++) {
for(int j = n; j > 1; j /= 2) {
System.out.println(j);
}
}
```

Question 9

```
for(int i = 1; i <= n; i++) {
for(int j = 1; j <= i; j++) {
System.out.println(j);
}
}
```

Question 10

```
for(int i = 0; i < n; i++) {
for(int j = 1; j < n; j *= 2) {}
for(int k = 0; k < n; k++) {}
}
```

Question 11

```
void fun(int n) {
if(n <= 1) return;
fun(n / 2);
}
```

Question 12

```
void fun(int n) {
if(n <= 1) return;
fun(n - 1);
fun(n - 1);
}
```

Question 13

```
HashMap map = new HashMap<>();
for(int i = 0; i < n; i++) {
map.put(i, i);
}
```

Question 14

```
void fun(int n) {
if(n <= 1) return;
fun(n / 2);
fun(n / 2);
}
```

Question 15

```
void permute(String str, String ans) {
if(str.length() == 0) {
System.out.println(ans);
return;
}
for(int i = 0; i < str.length(); i++) {
char ch = str.charAt(i);
String left = str.substring(0, i);
String right = str.substring(i + 1);
permute(left + right, ans + ch);
}
}
```